

Python
Type Analyzer

RQ1
(Analysis Approach)



- Static
- Dynamic
- Hybrid(Static + Dynamic)

RQ2
(Analysis Features)



- Structural Typing
- Flow-sensitivity
- Dynamic Features
- Compound DataTypes

RQ3
(Existing tools)



- Static Analyzer
- Dynamic Checking
- Hybrid Analyzer
- LLM-based Inference

- Mypy, pyright ...
- Typeguard, icontact ...
- MonkeyType, CrossHair ...
- Type4Py, TypeT5 ...